

Basic Game Object Manipulation and Transformations

Objectives

- Understand the concept of game objects and their role in Unity.
- Learn basic game object manipulation techniques.
- Grasp the fundamentals of transforming game objects in Unity.

Content

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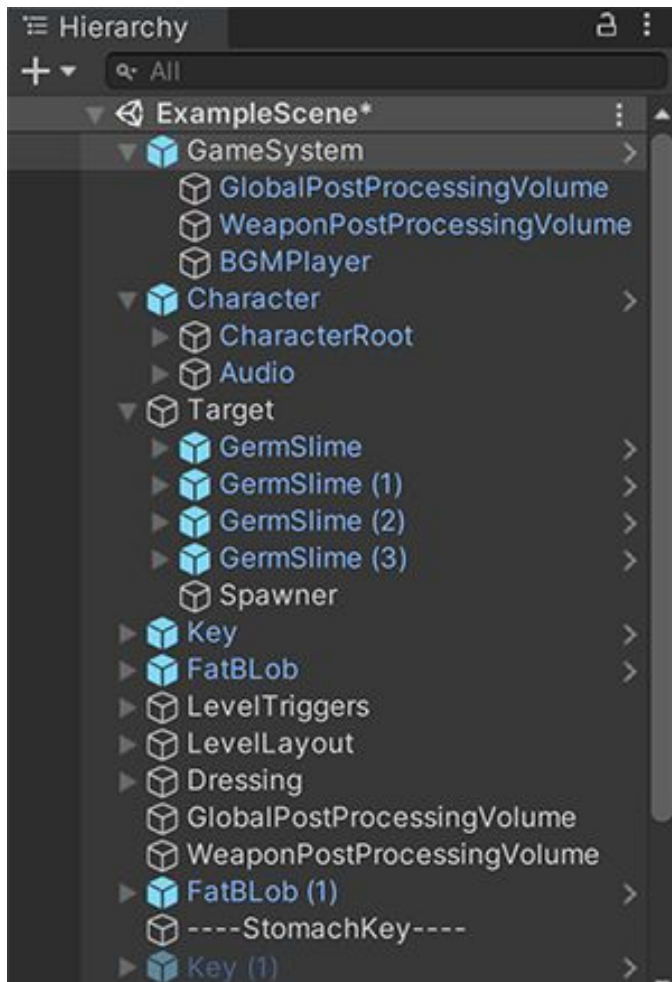
What are Game Objects?

GameObjects are the foundational components of a Unity game project.

“GameObjects are the fundamental objects in Unity that represent characters, props and scenery. They do not accomplish much in themselves but they act as containers for Components, which implement the functionality.”^[8]

Every object in your game is a GameObject, from characters and collectible items to lights, cameras and special effects.

Hierarchy Panel - GameObjects - Nested



A parent GameObject has other GameObjects connected to it that take on its Transform properties.

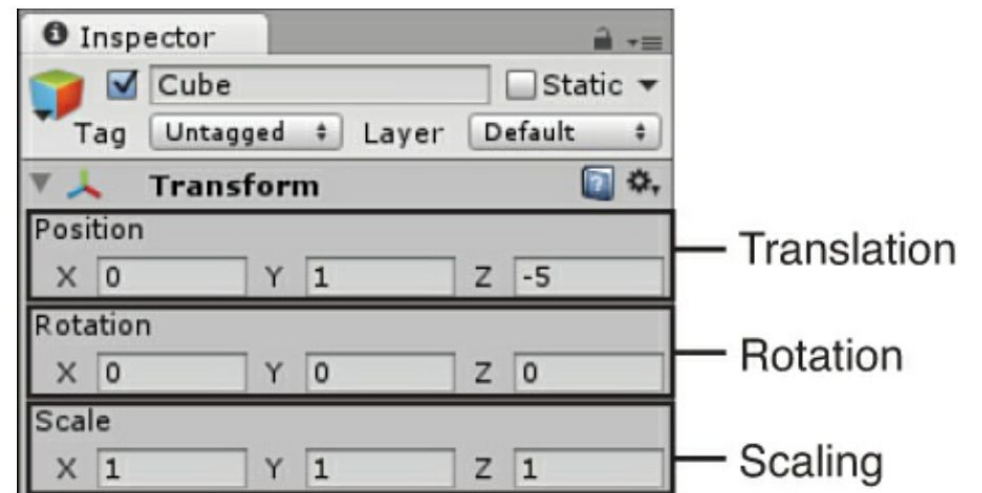
A child GameObject is connected to another GameObject, and takes on that GameObject's Transform properties.

A child GameObject moves, rotates, and scales exactly as its parent does. Child GameObjects can also have child GameObjects of their own. A GameObject can have multiple child GameObjects, but only one parent GameObject.

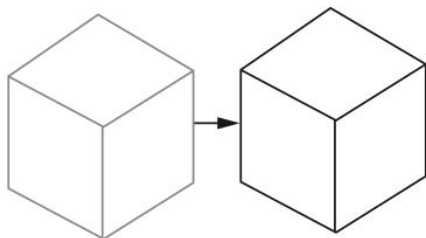
Transforms

“The Transform stores a GameObject’s Position, Rotation, Scale and parenting state. A GameObject always has a Transform component attached: you can’t remove a Transform or create a GameObject without a Transform component.”^[8]

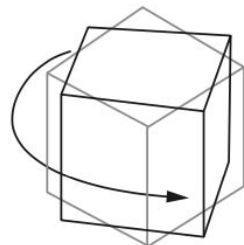
Property:	Function:
Position	Position of the Transform in the x, y, and z coordinates.
Rotation	Rotation of the Transform around the x-axis, y-axis, and z-axis, measured in degrees.
Scale	Scale of the Transform along the x-axis, y-axis, and z-axis. The value “1” is the original size (the size at which you imported the GameObject).



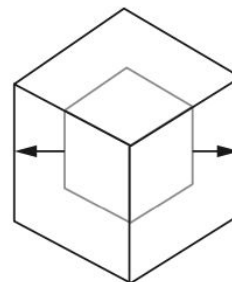
Translate



Rotate



Scale



Transform Tool

You can use the Transform tools on any GameObject in a scene.

When you select a GameObject, the tool Gizmo appears within it. The appearance of the Gizmo depends on which tool you select.

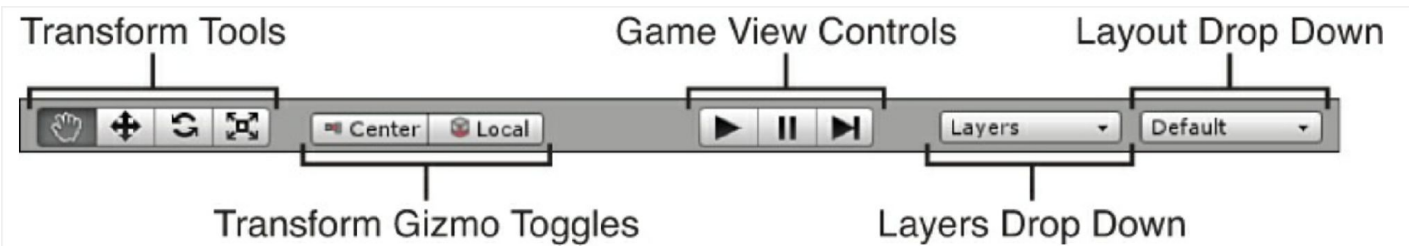
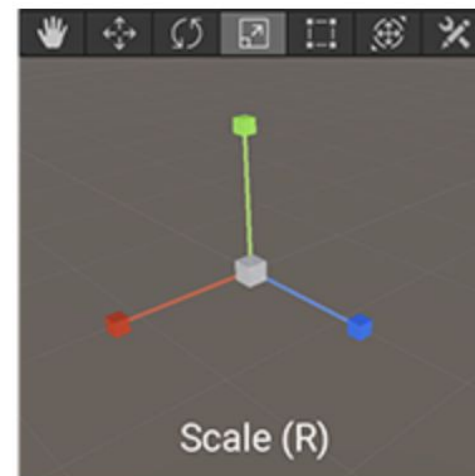
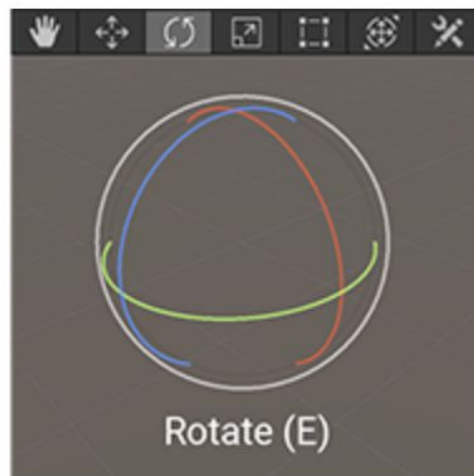


FIGURE 1.15 The toolbar.



Transforms - Understanding 3D coordinate space

Vertical axis
(usually labeled y)

Coordinates that define the point's
position. The numbers indicate each
distance along one axis: (x, y).

(6, 5)

Horizontal axis
(labeled x)

Figure 2.3 Coordinates along the x- and y-axes define a 2D point.

Vertical axis
(labeled y)

(6, 7, 5)

The z-axis is
perpendicular
to the page;
imagine this line
sticking straight
into and out
of the page.

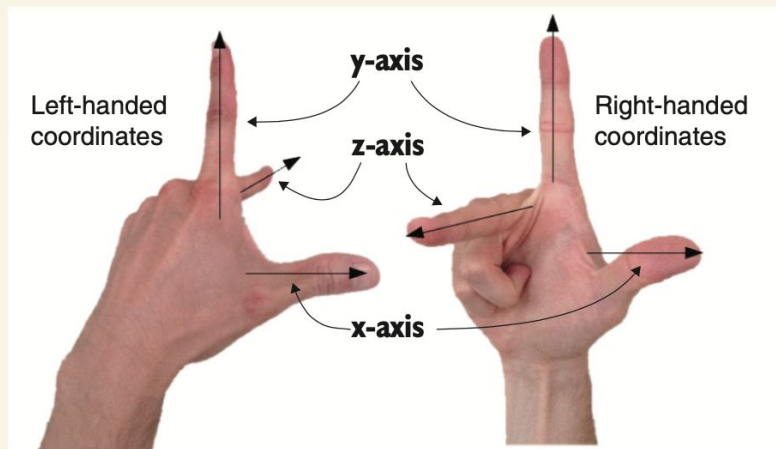
Whereas 2D coordinates have
two numbers, one along each
axis, 3D coordinates have
three numbers: (x, y, z).

Horizontal axis
(labeled x)

Figure 2.4 Coordinates along the x-, y-, and z-axes define a 3D point.

Transforms - Understanding 3D coordinate space cont

But in almost all cases, x goes to the right and y goes up; what differs between different tools is whether z goes into or comes out of the page. These two directions are referred to as *left-handed* or *right-handed*; as this figure shows, if you point your thumb along the x-axis and your index finger along the y-axis, then your middle finger points along the z-axis.



The z-axis points in a different direction on the left hand versus the right hand.

Unity uses a left-handed coordinate system, as do many 3D art applications. Many other tools use right-handed coordinate systems (OpenGL, for example), so don't get confused if you ever see different coordinate directions.

Transform Component in Inspector

At the top, you can type in a name for the object. For example, call the floor object `Floor`.

Position and scale the cube to create a floor for the room. It won't look like a cube anymore after being stretched out with differing scale values on different axes.

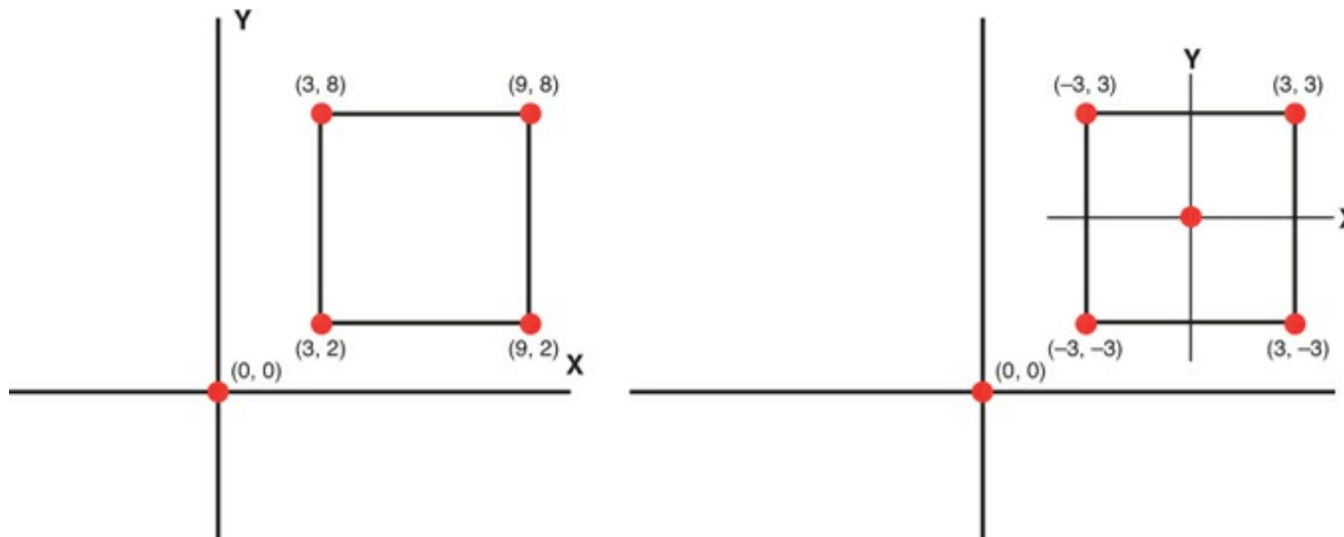
Meanwhile, the position is lowered slightly to compensate for the height; we set the Y scale to 1, and the object is positioned around its center.



The remaining components filling the view come with a new Cube object but don't need to be adjusted right now. These components include a Mesh Filter (to define the geometry of the object), a Mesh Renderer (to define the material on the object), and a Box Collider (so that the object can be collided with during movement).

Understanding local vs. world coordinate space

- Every game object in Unity has a position, but a position needs a reference frame. In particular, Unity offers two ways to see (and set) coordinates:
- World coordinates:** These are absolute coordinates of where the game object is located (by absolute, I mean with respect to the world frame, which is considered to be absolute in the game)
- Local coordinates:** These are the coordinates of where the game object is with respect to its parent



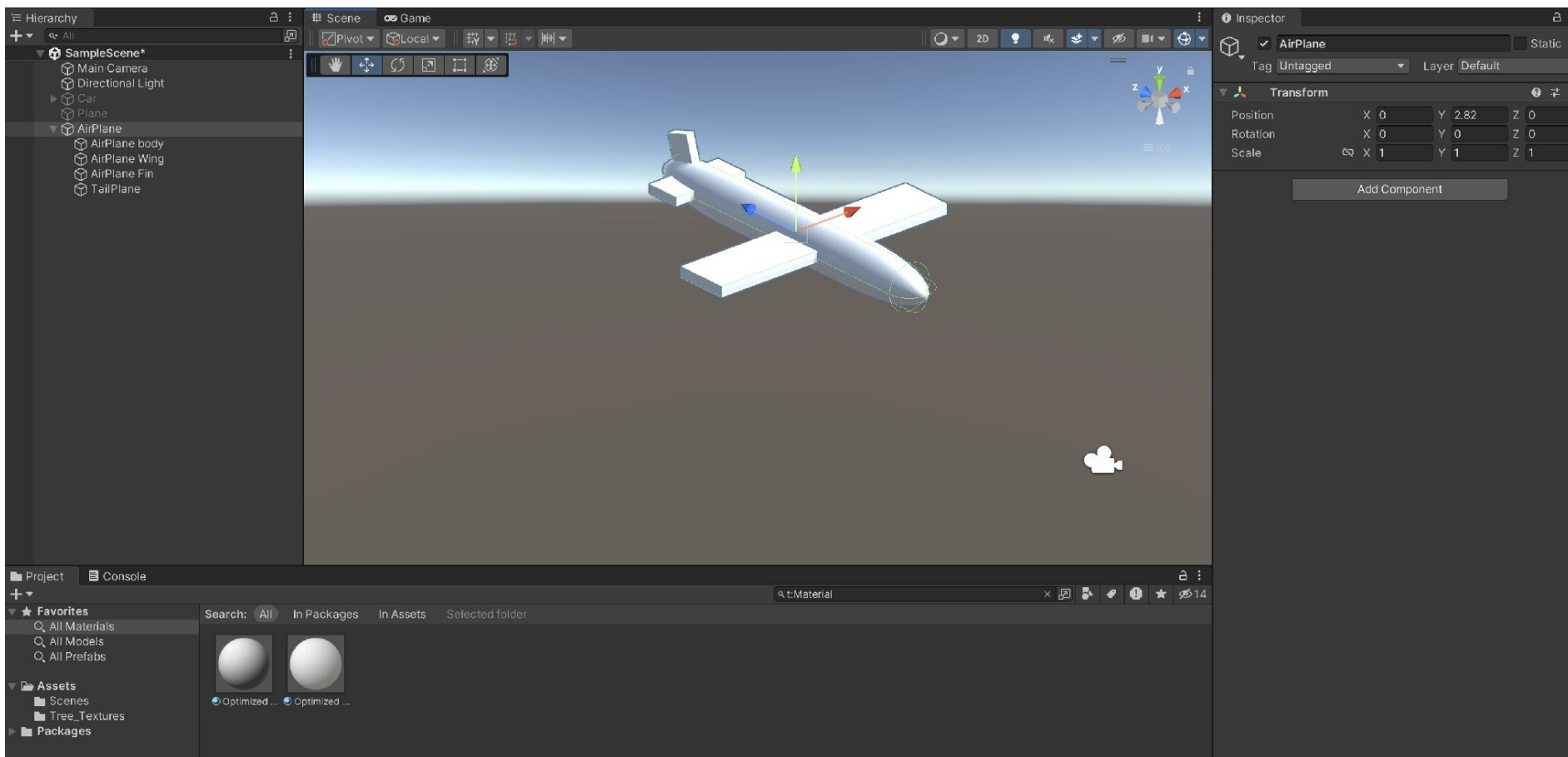
Parenting and Child Relationships

“Game objects can be the child of other game objects. When a game object moves, rotates, or scales, its children are affected as well. This means that you can create hierarchies of game objects that work together as a system; for example, a car could have each of its four wheels as child objects, which means that they automatically stay in the correct position as the car moves around.”^[2]

TIPs: Before linking any child objects to it, make sure to reset the Transform options (Position and Rotation to 0, 0, 0 and Scale to 1, 1, 1) of the empty root object to avoid any oddities in the position of child objects.

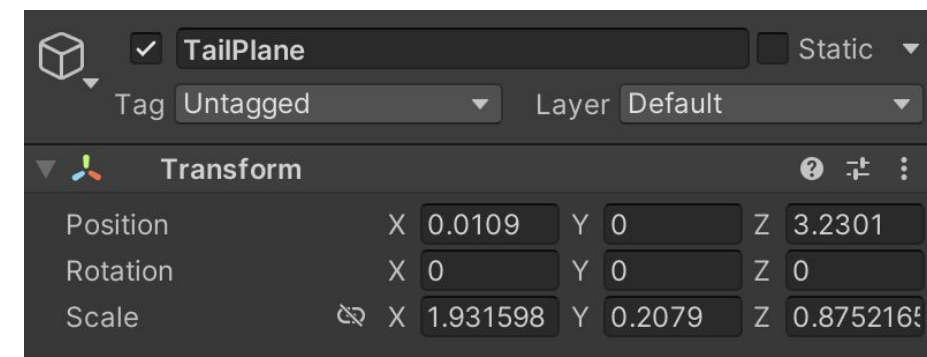
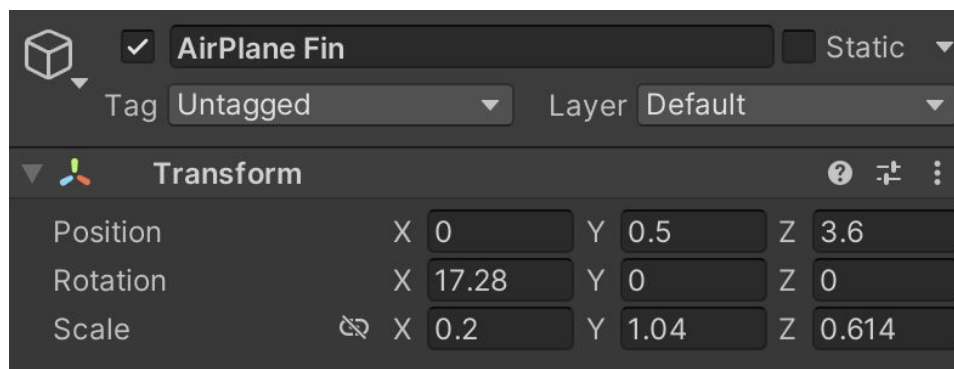
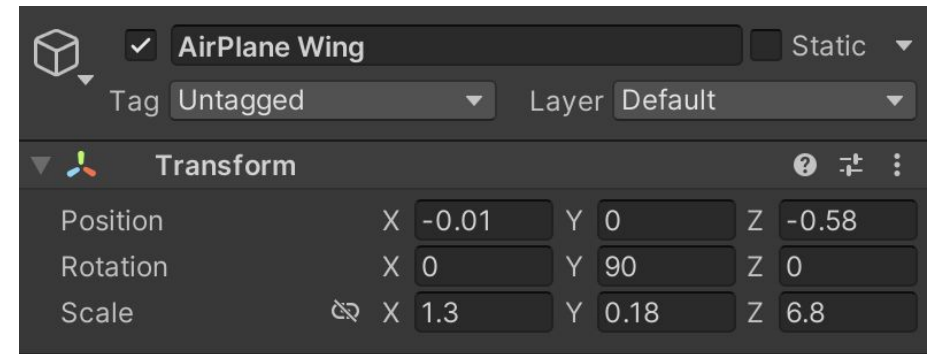
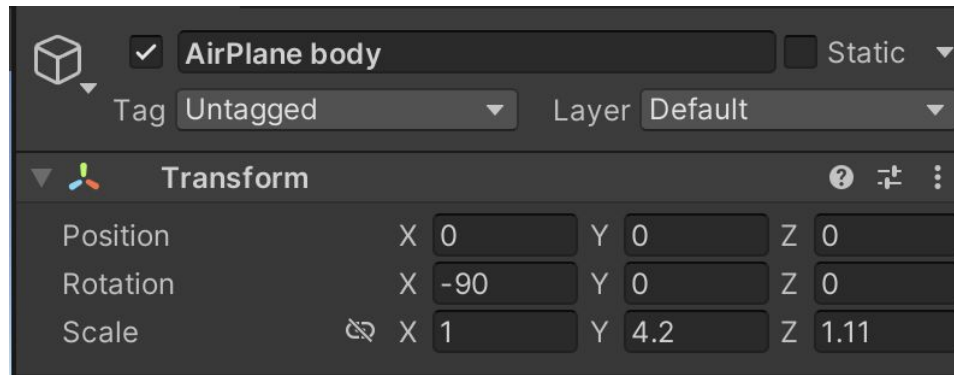
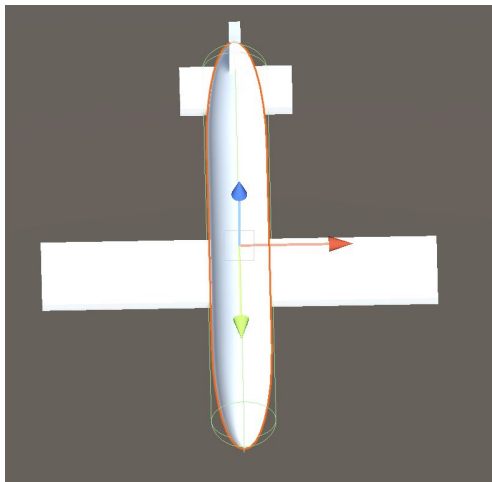
Pratice

In order to understand the moving of GameObject's transform, use Unity 3D for this practice.



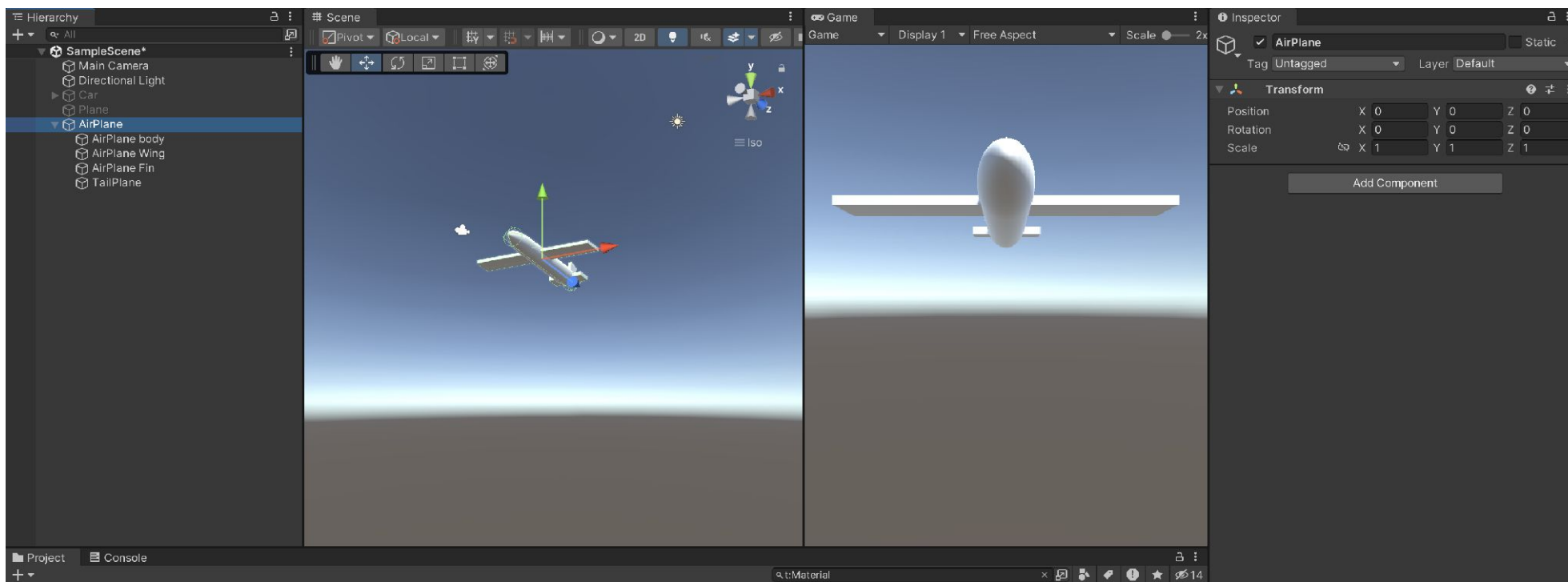
Pratice – Use transform tool to create 3D Objects

1. Add Capcule 3D GameObject. Rotate, scale it to make plane body.
2. Add Cube. Use React, scale, rotate and move tool to make airplane's wing
3. Add two Cube. Use React, scale, rotate and move tool to make airplane's fin and tailplane



Create nested object

1. Create empty 3D GameObject. Make sure the Position and Rotation values is 0 and Scale values is 1. Rename this empty object to AirPlane
2. Drop all airplane parts to AirPlane.
3. Move, rotate and scale AirPlane object to check the transform of other airplane's part
4. Reset Transform value of Airplane.



Conclusions

- Explored game object fundamentals and their significance in Unity scenes.
- Learned manipulation techniques—selecting, moving, rotating, and scaling objects.
- Mastered Transform component usage and understood coordinate systems' impact.
- Grasped parent-child relationships' influence on object transformations.

References

- 1: Hocking, Joseph; Schell, Jesse, Unity in action: multiplatform game development in C#, 2022
- 2: Buttfield-Addison Paris, Manning Jon, Nugent Tim, Unity game development cookbook: essentials for every game, 2019
- 3: Geig, Mike, Sams teach yourself Unity Game development in 24 hours, 2014
- 8: Unity Technologies, Unity Manual, 2023